VF5b caustics: softer, uneven, slow-machine safeguards

Look (your notes): weaker and softer (spot 0.85 -> 0.6, LED bar 0.55 -> 0.3, tube 0.4 -> 0.15, more blur under bars and tubes), a 30 % larger, slower net; fades in the shadow of rock, wood and plant canopies on the sand; steep faces drop off fast; broad slow patches of stronger / weaker shimmer, so it is not one even overlay. LED tanks: subtle. Spot (reef): clearest.

Safeguards: moves only at the Full quality performance level; stops after 1 minute with no mouse / key input (starts again on the next move); on a slow machine it halves its rate, then holds the pattern still (still = same look, no redraws).

Measured on this PC, 15 s each (redraws per second, GPU busy, CPU busy as % of one core):

- Caustics off, tank left alone: 0 redraws, nothing busy (as before).
- Caustics on, tank left alone: 18 redraws/s; GPU 12 % (nano) to 23 % (planted, reef); CPU 5-8 %.
- Orbiting: 60 fps with or without caustics; slowest frames unchanged (about 18-19 ms).
- Caustics on, no input for a minute: 0 redraws.
- CPU slowed 4x (stand-in for a weak laptop), left alone: all three tanks stop the motion within 2 s (then 0 cost).

Not covered: a truly weak GPU (the browser can only slow the CPU) and phones; the back-off watches frame delays for that.

Pages: No caustics | First version (you saw it earlier; before VF-C's ground and contact work) | Tuned (now).

Planted 4 ft (LED bar) - front view



No caustics | **First version (you saw this)** | **Tuned (now)**

Planted 4 ft (LED bar) - ground view



No caustics | **First version (you saw this)** | **Tuned (now)**

Planted 4 ft (LED bar) - rock view



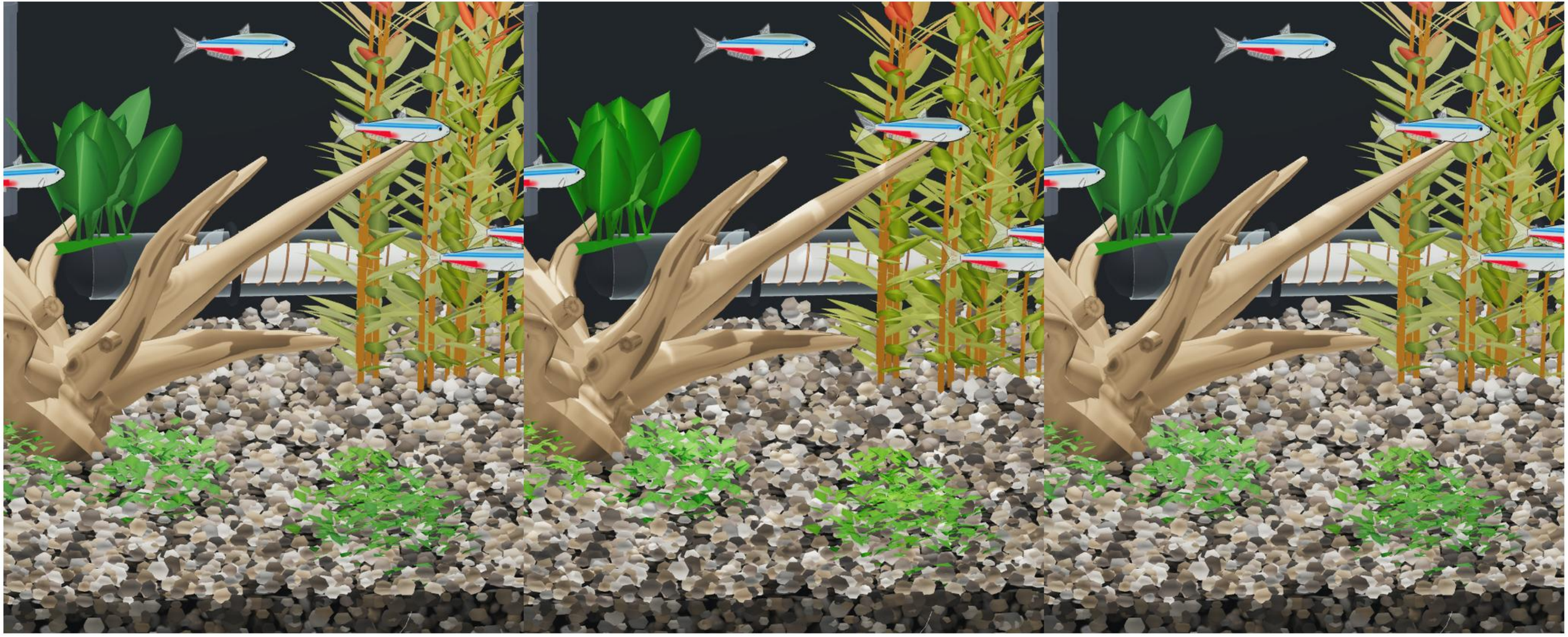
No caustics | **First version (you saw this)** | **Tuned (now)**

Rimless nano (LED bar) - front view



No caustics | **First version (you saw this)** | **Tuned (now)**

Rimless nano (LED bar) - ground view



No caustics | **First version (you saw this)** | **Tuned (now)**

Rimless nano (LED bar) - rock view



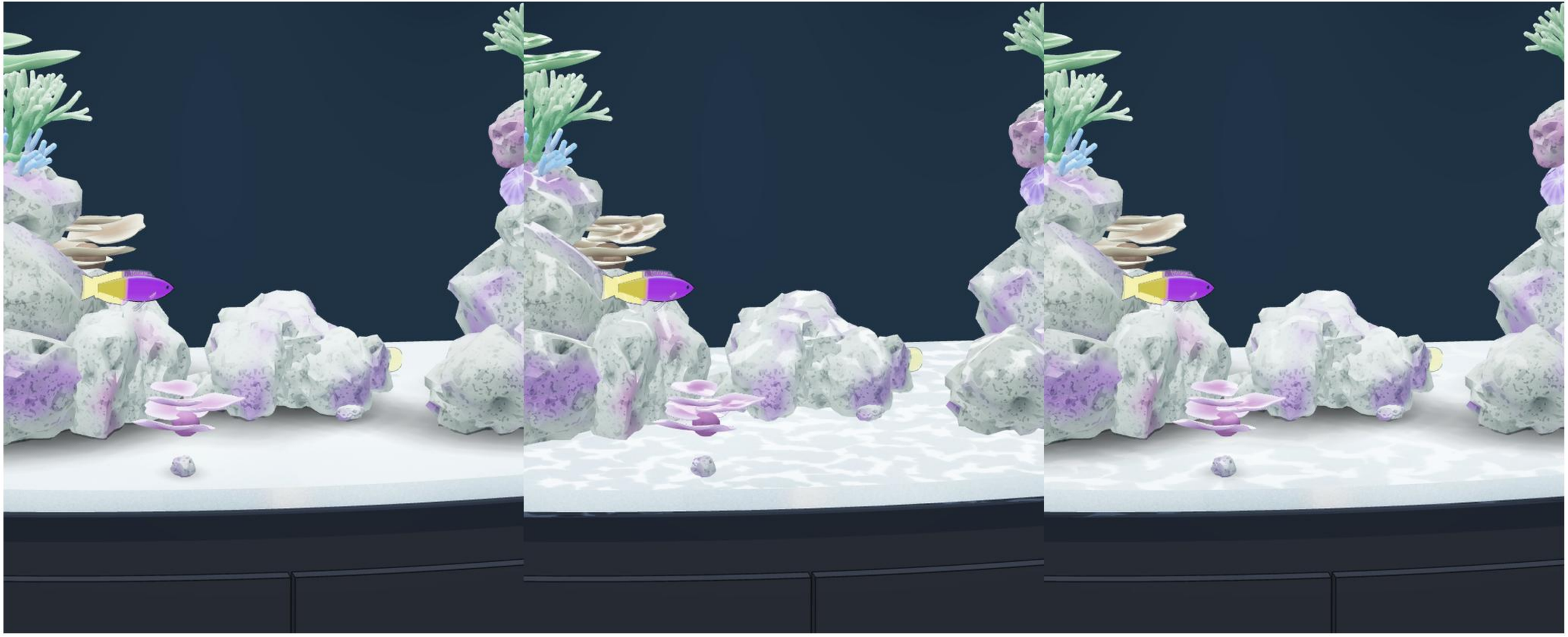
No caustics | **First version (you saw this)** | **Tuned (now)**

Reef bow-front (spots) - front view



No caustics | **First version (you saw this)** | **Tuned (now)**

Reef bow-front (spots) - ground view



No caustics | **First version (you saw this)** | **Tuned (now)**

Reef bow-front (spots) - rock view



No caustics | **First version (you saw this)** | **Tuned (now)**